

A uniform pole-free disk for cubic Ihara zeta functions

Disproof of Conjecture 00000007915

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Abstract

For every finite cubic simple graph, all Ihara poles have modulus at least $1/2$. The circle proposed in the source has radius $1/(2\sqrt{2}) < 1/2$. A fixed positive gap separates every pole of every such graph from that circle, ruling out the asserted equidistribution even after only nontrivial poles are retained.

The asserted circle and the actual operator

For $k = 3$, the source proposes pole equidistribution on

$$S_r = \{u \in \mathbb{C} : |u| = r\}, \quad r = \frac{1}{2\sqrt{2}}.$$

We use the unrescaled Ihara variable u appearing in that assertion. Let $D(G)$ be the oriented edges of a finite simple graph. Define its nonbacktracking matrix by

$$B_{(a,b),(c,d)} = \begin{cases} 1, & b = c \text{ and } d \neq a, \\ 0, & \text{otherwise.} \end{cases}$$

If G is cubic, each row has exactly two entries equal to 1: at the current endpoint there are three neighbors, of which immediate reversal excludes one. In particular, every row sum is at most 2. The Hashimoto formula stated as the framework of the source is

$$\zeta_G(u) = \frac{1}{P_G(u)}, \quad P_G(u) = \det(I - uB).$$

This identifies the operator we use with the genuine Ihara function. Because the numerator is 1 and $P_G(0) = 1$, its finite poles are exactly the zeros of P_G , counted with multiplicity. The determinant formula is also recorded, for example, in¹ the standard graph-zeta literature.

A bound uniform over all cubic graphs

Lemma 1. *For every finite cubic simple graph G , $P_G(u) \neq 0$ whenever $|u| < 1/2$.*

Proof. If $P_G(u) = 0$, there is a nonzero vector $x \in \mathbb{C}^{D(G)}$ such that $(I - uB)x = 0$, or $x = uBx$. Choose a coordinate a for which $M = |x_a| = \max_e |x_e| > 0$. The row bound gives

$$M = |u| |(Bx)_a| \leq |u| \sum_{e: B_{a,e}=1} |x_e| \leq 2|u|M.$$

Dividing by $M > 0$ yields $|u| \geq 1/2$, a contradiction. $\square$

Set $\delta = 1/2 - r > 0$. For every pole u and every $w \in S_r$, the reverse triangle inequality gives

$$|u - w| \geq |u| - |w| \geq \frac{1}{2} - r = \delta.$$

The same δ works for every graph size and every realization of a random cubic graph. These bounds apply to all poles, and hence also to any collection selected by nontrivial spectral parameters.

¹The Ihara Zeta function and Quantum Walk, Theorem 3 (Hashimoto–Bass).

Why this disproves equidistribution

This is a uniform obstruction, rather than an atypical finite example. Define the bounded continuous function

$$\varphi(z) = \max \left\{ 0, 1 - \frac{||z| - r|}{\delta} \right\}.$$

It equals 1 at every point of S_r and 0 at every possible pole, since those poles have modulus at least $1/2 = r + \delta$. Every normalized empirical pole measure therefore integrates φ to 0, as does its expectation. Any probability measure supported on S_r , in particular uniform angular measure, integrates φ to 1. Weak convergence to such a measure is impossible. The argument also rules out convergence in probability or almost surely for random empirical measures: the test integral is identically zero at every stage. Deleting trivial poles or changing multiplicities cannot produce mass in the separated neighborhood of the circle. Thus the first assertion of the source fails already at $k = 3$.

Correspondence with Lean

The matrix `nonbacktracking` is built on actual `SimpleGraph.Dart` objects. The theorem `successors_card_le` injects its legal successor set into the current vertex's neighbor set with the reverse endpoint erased; actual graph degree 3 supplies the bound 2. The proof of `ihara_denominator_nonzero` uses Mathlib's equivalence between zero matrix determinant and a nonzero kernel vector, then the actual maximum-coordinate argument above. The final theorem `uniform_separation_from_claimed_circle` gives the positive gap for every genuine denominator root. The complete graph K_4 formally witnesses that the cubic hypothesis is realizable.

The formal result is the universal graph-determinant obstruction. The Hashimoto identification of this denominator with Ihara zeta and the bounded-continuous-test definition of weak convergence are the explicit mathematical bridges supplied above. The Lean file does not claim to formalize the general Ihara Euler-product theorem or a random-graph limit theorem.

Reproduction

The project pins Lean 4.19.0 and Mathlib commit `c44e0c8ee63ca166450922a373c7409c5d26b00b`.

From `lean/`, run `lake build` and `lake env lean -DwarningAsError=true Main.lean`. On a new machine, `lake exe cache get` retrieves the official pinned dependency artifacts. The submission itself is checked in a fresh build directory. Printed dependencies use only the standard `propext`, `Classical.choice`, and `Quot.sound`. No additional axioms, proof placeholders, or numerical oracles are used. Run `tectonic main.tex` to reproduce the PDF; no auxiliary numerical script is needed. Build logs, source hashes, and visual-review records are included under `verification/`.

Conjecture source

The Last Math Competition, Conjecture 00000007915. The exact bilingual source and its commit and SHA-256 are preserved in this submission.